

# EE8 — Batch Diverter

5 items to Gate A, 3 items to Gate B, repeat

The batch diverter sends 5 items to Gate A, then 3 items to Gate B, alternating forever. It uses the full v4 platform: I/O ports (IN/OUT), the ITEM\_PULSE sensor latch, and the motor safety interlock. All 16 instruction slots are used.

## 1 I/O Port Bits

| Port   | Bit | Signal                                    |
|--------|-----|-------------------------------------------|
| Input  | 0   | ITEM_PULSE — set by sensor, cleared on IN |
| Output | 0   | GATE_SEL — 0 = Gate A, 1 = Gate B         |
| Output | 1   | MOTOR_EN — motor enable request           |

0x02 = 0b00000010: motor on, Gate A. 0x03 = 0b00000011: motor on, Gate B.

## 2 Register Allocation

| Register | Usage                                          |
|----------|------------------------------------------------|
| R0       | Items remaining in current batch (counts down) |
| R1       | Scratch — masks, constants                     |
| R2       | Output port value (written by OUT R2)          |
| R3       | Scratch — ITEM_PULSE mask, gate select test    |

## 3 Assembly Program

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1  ;           Batch Diverter: A=5, B=3, alternating
2  ;
3  ; Setup
4  0:  LDI  R0, 5           ; Batch count A
5  1:  LDI  R2, 0x02        ; MOTOR_EN=1, GATE_SEL=0 (Gate A)
6
7  ; Counting loop (shared for both gates)
8  2:  OUT  R2              ; Drive output port
9  3:  IN   R1              ; Read input (clears ITEM_PULSE)
10 4:  LDI  R3, 0x01        ; ITEM_PULSE mask
11 5:  AND  R1, R3, R1      ; Isolate bit 0
12 6:  JZ   R1, 3           ; No item? Keep polling
13 7:  LDI  R1, 1           ; Decrement constant
14 8:  SUB  R0, R1, R0      ; R0--
15 9:  JNZ  R0, 3           ; More items? Next item
16
17 ; Batch complete: toggle gate, reload count
18 10: LDI  R1, 0x01        ; GATE_SEL toggle mask
19 11: XOR  R2, R1, R2      ; Flip bit 0: 0x02 <-> 0x03
20 12: AND  R2, R1, R3      ; R3 = new GATE_SEL (0 or 1)
21 13: LDI  R0, 3           ; Load count B
22 14: JNZ  R3, 2           ; Gate B? Start counting
23 15: LDI  R0, 5           ; Gate A: load count A
24                ; PC wraps 15->0->1->2

```

**Note**

After Gate A completes, XOR flips R2 from 0x02 to 0x03. AND with 0x01 extracts the new GATE\_SEL into R3. If R3 = 1 (Gate B), JNZ jumps to address 2. If R3 = 0 (Gate A), execution falls through to address 15 which loads count A, then the PC wraps to address 0 (redundant LDI R0,5) and address 1 (restores R2 = 0x02).

## 4 Machine Code

| Addr | Hex  | Instruction    |
|------|------|----------------|
| 0    | A005 | LDI R0, 5      |
| 1    | A802 | LDI R2, 0x02   |
| 2    | 7A00 | OUT R2         |
| 3    | 7340 | IN R1          |
| 4    | AC01 | LDI R3, 0x01   |
| 5    | 2740 | AND R1, R3, R1 |
| 6    | C403 | JZ R1, 3       |
| 7    | A401 | LDI R1, 1      |
| 8    | 1100 | SUB R0, R1, R0 |
| 9    | D003 | JNZ R0, 3      |
| A    | A401 | LDI R1, 0x01   |
| B    | 4980 | XOR R2, R1, R2 |
| C    | 29C0 | AND R2, R1, R3 |
| D    | A003 | LDI R0, 3      |
| E    | DC02 | JNZ R3, 2      |
| F    | A005 | LDI R0, 5      |

## 5 Execution Summary

| Phase   | Addresses          | R0  | R2   | Output           |
|---------|--------------------|-----|------|------------------|
| Setup   | 0→1                | 5   | 0x02 | Motor on, Gate A |
| Batch 1 | 2→...→9 (5 items)  | 5→0 | 0x02 | Gate A           |
| Toggle  | A→B→C→D→E→2        | 3   | 0x03 | Gate B           |
| Batch 2 | 2→...→9 (3 items)  | 3→0 | 0x03 | Gate B           |
| Toggle  | A→...→F→0→1→2      | 5   | 0x02 | Gate A           |
| Batch 3 | repeats as Batch 1 |     |      |                  |

Pattern: 5(A), 3(B), 5(A), 3(B), ... forever.